Climate Change

Understanding Causes, Impacts, and Solutions

NAME -: PREREAK KALPESHBHAI THAKKAR

ENROLLMENT NUMBER -: 22C21120

B.TECH CSE-2

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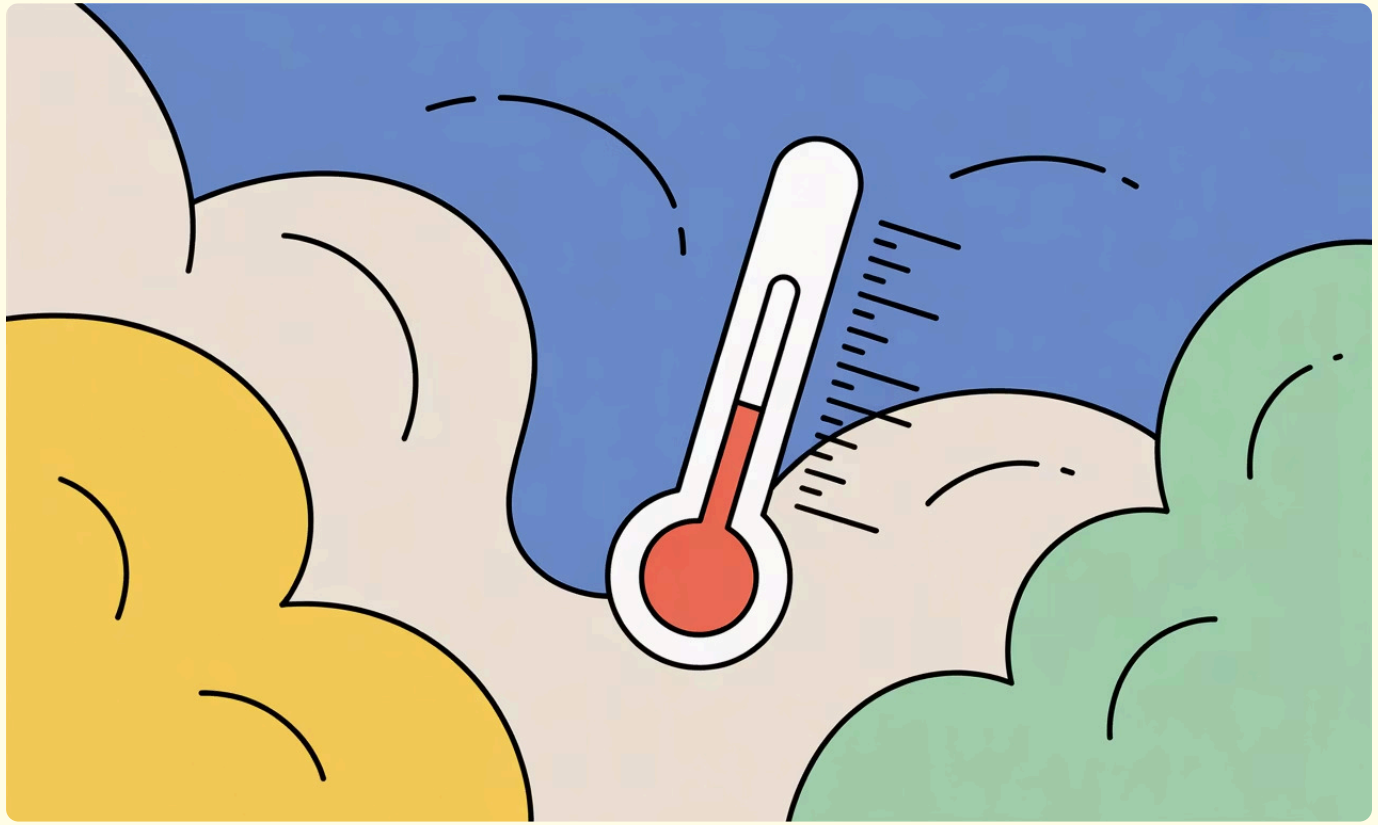
What is Climate Change?



Climate change refers to long-term shifts in temperatures and weather patterns. Whilst these changes can occur naturally through volcanic eruptions or solar cycles, human activities have become the primary driver since the industrial revolution.

Today, climate change affects ecosystems, economies, and communities across every continent, reshaping our planet in unprecedented ways.

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Key Indicators of Climate Change

Global Warming

Steady increase in Earth's average surface temperature, with the warmest years on record occurring in the past decade.

Rainfall Shifts

Significant changes in precipitation patterns, causing droughts in some regions whilst others experience unprecedented flooding.

Extreme Weather

More frequent and intense hurricanes, heatwaves, wildfires, and severe storms disrupting communities worldwide.

Climate Systems

Long-term disruption of established weather patterns, ocean currents, and seasonal cycles that life depends upon.

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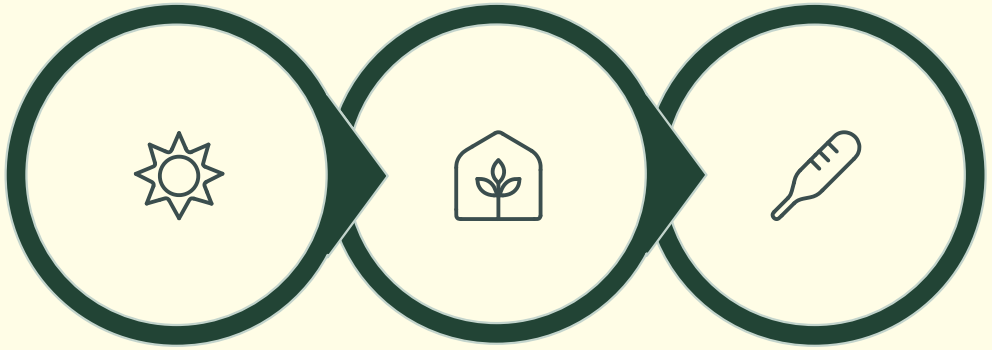
The Greenhouse Effect



This natural process has sustained life on Earth for millions of years by maintaining a habitable temperature.

Nature's Thermal Blanket

The greenhouse effect is Earth's natural warming system. Greenhouse gases in our atmosphere act like a blanket, trapping heat from the sun and keeping our planet warm enough to support life.

However, human activities have dramatically increased these gases, thickening the blanket and causing Earth to overheat—disrupting the delicate balance that sustains our ecosystems.

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Major Greenhouse Gases

Carbon Dioxide (CO_2)

The most abundant greenhouse gas from human activity, primarily released through burning fossil fuels and deforestation. CO_2 persists in the atmosphere for centuries.

Methane (CH_4)

A potent greenhouse gas released from agriculture, livestock digestion, and natural gas extraction. Methane traps significantly more heat than CO_2 over short periods.

Nitrous Oxide (N_2O)

Emitted from agricultural fertilisers, industrial processes, and combustion. Though less abundant, it's nearly 300 times more effective at trapping heat than CO_2.

Fluorinated Gases

Synthetic gases used in refrigeration, air conditioning, and industrial applications. Extremely powerful heat trappers, though present in smaller quantities.

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Human Causes of Climate Change

Burning Fossil Fuels

Coal, oil, and natural gas combustion for electricity, heating, and power releases billions of tonnes of CO_2 annually—the single largest contributor to climate change.

Deforestation

Clearing forests for agriculture and development eliminates vital carbon sinks. Trees that once absorbed CO_2 are destroyed, releasing stored carbon back into the atmosphere.

Industrial Activities

Manufacturing, cement production, and chemical processes emit substantial greenhouse gases whilst also consuming enormous amounts of fossil fuel energy.

Agriculture & Livestock

Farming practices, particularly livestock raising, generate significant methane and nitrous oxide. Rice paddies and manure management contribute substantially to emissions.

Transportation

Vehicles, aeroplanes, and ships burning petrol and diesel account for nearly a quarter of global CO_2 emissions, with numbers rising as transportation demand grows.

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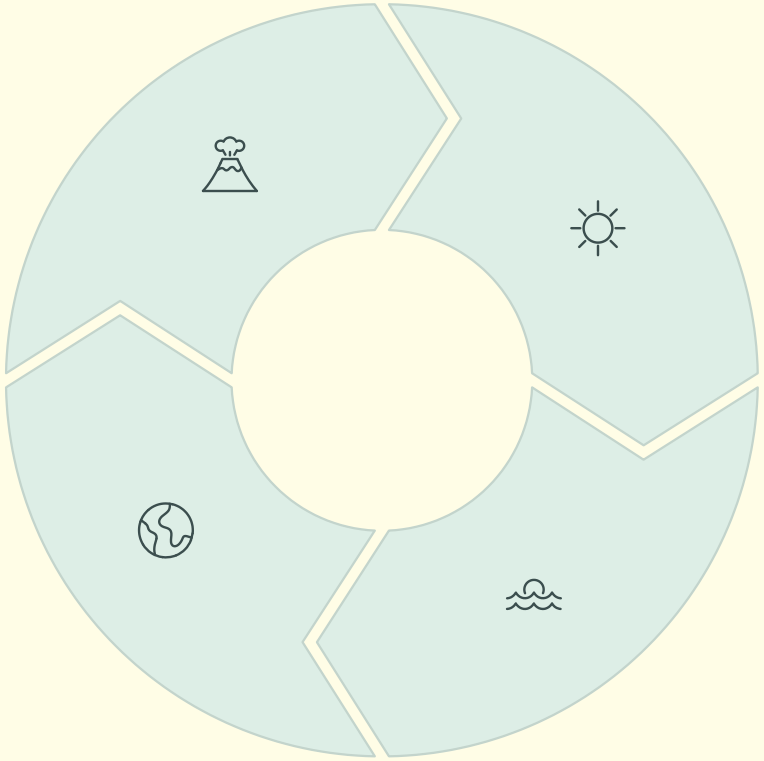
Natural Climate Influences

Whilst human activity is the primary driver today, natural processes have shaped Earth's climate throughout history.



Volcanic Eruptions

Release gases and particles that can temporarily cool or warm the planet.

Solar Radiation

Variations in the sun's energy output affect Earth's temperature over long cycles.

Natural Cycles

Earth's orbital changes create ice ages and warming periods over millennia.

Ocean Currents

Shifting patterns in ocean circulation redistribute heat across the globe.

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Evidence of Climate Change

Scientific Observations

Multiple lines of evidence from around the world paint an unmistakable picture of a rapidly changing climate. Satellites, ocean buoys, weather stations, and ice cores all tell the same story.

The science is clear and the data is overwhelming: our planet is warming at an unprecedented rate, with consequences visible across every ecosystem and continent.

Rising Global Temperatures

Average global temperature has increased by 1.1°C since pre-industrial times, with accelerating warming in recent decades.

Melting Ice

Glaciers and polar ice caps are retreating at alarming rates, with Arctic sea ice extent declining by 13% per decade.

Rising Sea Levels

Oceans have risen approximately 20cm since 1900, threatening coastal communities and island nations worldwide.

Wildlife Changes

Species are migrating to cooler regions, breeding seasons are shifting, and coral reefs are bleaching from heat stress.

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CRITICAL ISSUE

Global Warming: The Core Challenge

Continuous Temperature Rise

Earth's surface temperature is rising persistently, with each of the last four decades warmer than any decade since 1850.

CO_2 as Main Driver

Atmospheric CO_2 concentrations have reached 420 parts per million—50% higher than pre-industrial levels—primarily from fossil fuel combustion.

Long-term Imbalance

Without urgent action, we risk triggering irreversible tipping points that would fundamentally alter Earth's climate system for generations.

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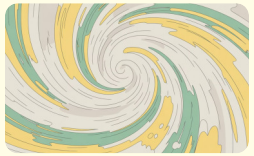
Extreme Weather Impact

Climate change is intensifying weather extremes, making once-rare events increasingly common and severe.

Devastating Heatwaves

Record-breaking temperatures are occurring more frequently, causing heat-related illnesses, wildfires, and strain on power grids as vulnerable populations struggle to cope.



Stronger Storms

Hurricanes and cyclones are intensifying with warmer ocean waters, bringing stronger winds, heavier rainfall, and more destructive storm surges to coastal regions.

Irregular Rainfall

Precipitation patterns are becoming more erratic—some areas face unprecedented droughts whilst others experience catastrophic flooding from intense downpours.

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The Urgent Reality of Climate Change

Understanding the profound impacts on our environment, health, and economy—and what we can do about it

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Environmental Devastation

Loss of Biodiversity

Species extinction rates have accelerated dramatically, with ecosystems collapsing as temperatures rise and habitats vanish

Coral Reef Bleaching

Ocean warming has devastated coral reefs worldwide, threatening marine life and coastal communities dependent on these vital ecosystems

Forest Fires

Increasingly severe wildfires consume vast areas, releasing massive carbon stores and destroying irreplaceable natural habitats

Desertification

Fertile lands transform into barren deserts, displacing communities and eliminating agricultural productivity across entire regions

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Human Health Under Threat

Heat-Related Illnesses

Record-breaking temperatures cause heat exhaustion, heatstroke, and cardiovascular stress, particularly affecting vulnerable populations

Infectious Diseases

Warmer climates expand the range of disease-carrying insects, spreading malaria, dengue fever, and other vector-borne illnesses

Food and Water Scarcity

Droughts and changing rainfall patterns threaten water supplies and agricultural yields, leading to malnutrition and conflict

Poor Air Quality

Increased wildfires and pollution exacerbate respiratory conditions, causing asthma, lung disease, and premature deaths

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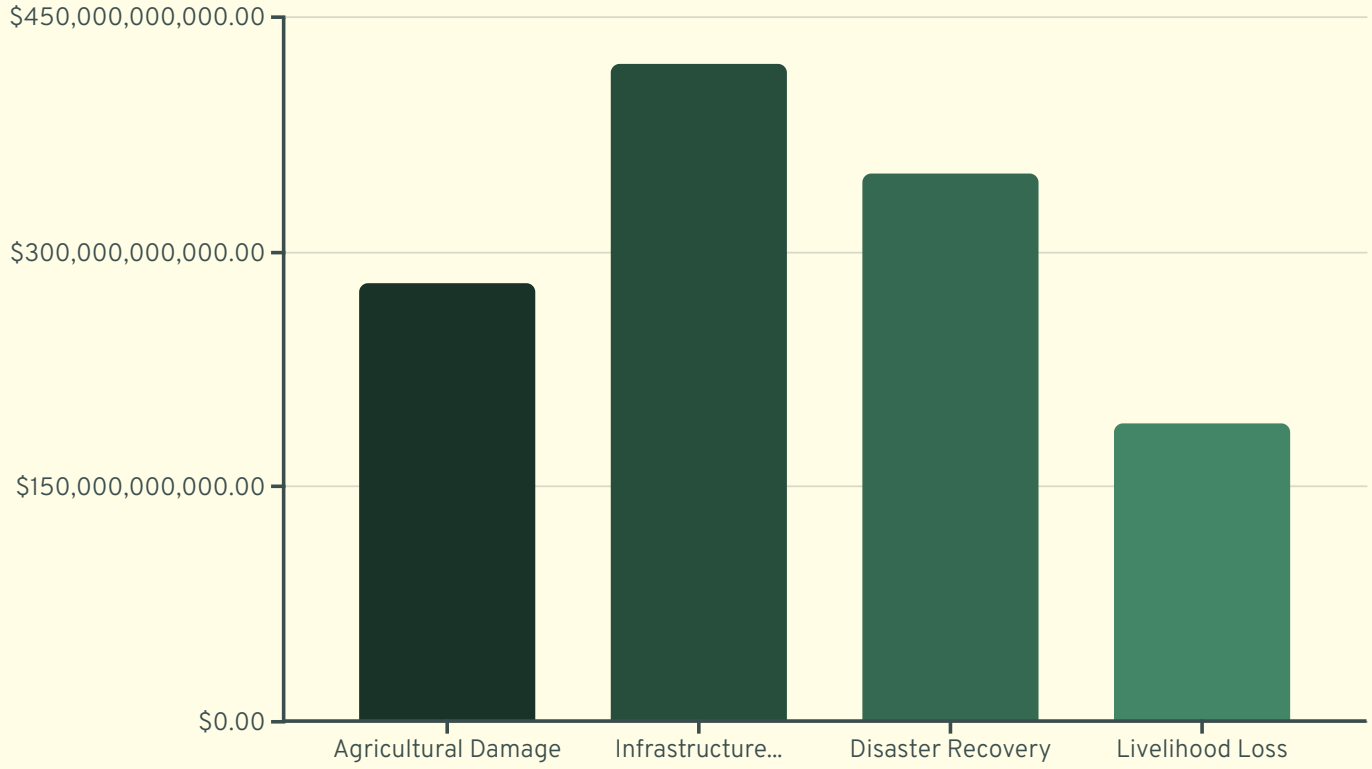
Economic Consequences



The global economy faces staggering losses from climate change. Crop failures devastate farming communities, whilst extreme weather destroys homes, roads, and vital infrastructure.

Disaster recovery costs spiral upwards each year, draining resources that could have been invested in prevention and adaptation. Millions lose their livelihoods as traditional industries collapse under environmental pressure.

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Climate Change in India

Rising Temperatures

India experiences increasingly severe heatwaves, with some regions recording temperatures exceeding 50°C, threatening lives and livelihoods

Erratic Monsoons

Unpredictable rainfall patterns disrupt agriculture, with intense downpours alternating with prolonged dry spells affecting millions of farmers

Floods and Droughts

Extreme weather events intensify, causing devastating floods in some areas whilst other regions face catastrophic water shortages

Coastal Threats

Rising sea levels endanger coastal communities, threatening major cities like Mumbai and millions living in vulnerable low-lying areas

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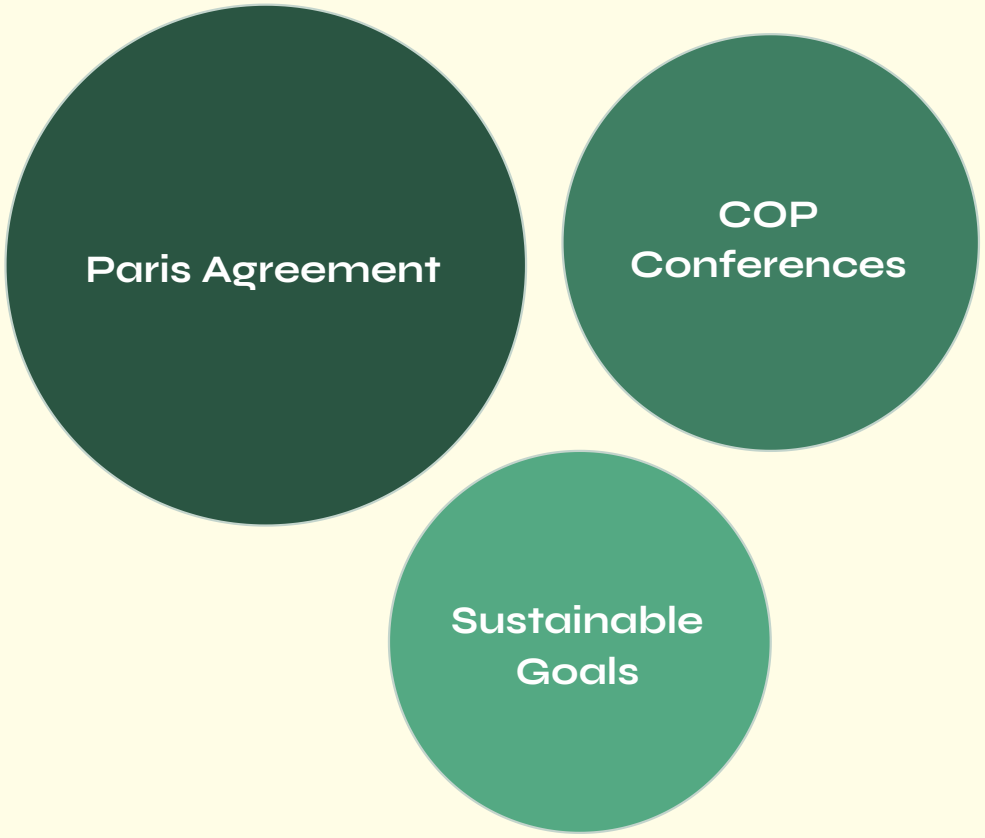
Global Efforts to Combat Climate Change



International cooperation has never been more critical in addressing the climate crisis facing our planet

United in Action

The Paris Agreement brought 196 nations together with a shared commitment to limit global temperature rise to well below 2°C, preferably to 1.5°C above pre-industrial levels.

Annual UN Climate Conferences (COP) provide platforms for countries to review progress, negotiate new commitments, and strengthen collective climate action.

The Sustainable Development Goals integrate climate action into broader development objectives, recognising that environmental sustainability underpins all progress.

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Renewable Energy Solutions

Solar Energy

Harnessing the sun's abundant power through photovoltaic panels and solar thermal systems, providing clean electricity for homes, businesses, and entire communities

Wind Energy

Capturing kinetic energy from wind using turbines, both onshore and offshore, generating substantial emissions-free power across the globe

Hydropower

Utilising flowing water to generate reliable, renewable electricity whilst supporting water management and flood control in many regions

Bioenergy

Converting organic materials into sustainable fuels and energy, reducing waste whilst providing alternatives to fossil fuels for heating and transport

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Individual Actions That Matter

Every Choice Counts

Reduce Energy Consumption

Switch to energy-efficient appliances, insulate homes properly, and adopt mindful consumption habits

Use Public Transport

Choose buses, trains, or cycling over private vehicles to dramatically reduce your carbon footprint

Save Resources

Conserve electricity and water through simple daily actions—turn off lights, fix leaks, use efficiently

Reduce, Reuse, Recycle

Minimise waste, repair rather than replace, and recycle materials to create a circular economy

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Leadership from Governments and Industries

Implement Climate Policies

Establish binding legislation with measurable targets for emissions reduction and environmental protection

Promote Clean Energy

Invest in renewable infrastructure and provide incentives for businesses and citizens to transition away from fossil fuels

Enforce Environmental Laws

Hold polluters accountable through robust regulation, monitoring, and meaningful penalties for environmental damage

Support Green Technologies

Fund research, development, and deployment of sustainable innovations that accelerate the transition to a low-carbon economy

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The Time to Act Is Now

> Climate change is the defining challenge of our generation. Every action we take today—individually, collectively, and globally—shapes the world we leave for future generations.

Individual Responsibility

Make sustainable choices in daily life

Collective Action

Demand accountability from leaders

Global Cooperation

Support international climate agreements

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Future Outlook: Urgency and Opportunity

1	Immediate, decisive action can significantly reduce climate damage, as highlighted by the International Energy Agency's 2025 report emphasising near-term preparedness and emissions cuts.
2	Sustainable development is essential: transitioning energy systems to renewables like solar and wind, improving energy efficiency, and adopting emerging technologies such as AI and blockchain for climate solutions.
3	Collective global responsibility is critical—no single country can solve this alone; international frameworks like the Paris Agreement and UN Sustainable Development Goal 13 guide coordinated efforts to limit warming to 1.5°C.

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Conclusion: The Climate Challenge and Our Path Forward

The Global Crisis

Climate change is a serious, accelerating global crisis, with 2024 recorded as the hottest year at 1.55°C above pre-industrial levels (WMO).

Human Impact & Solutions

Human activities remain the primary driver, but science-based solutions exist today—from decarbonising energy to protecting ecosystems and innovating low-carbon technologies.

Urgent Action Required

The window to act is closing fast; urgent, large-scale deployment of these solutions is needed to avoid catastrophic impacts.

Collective Protection

Together, through policy, innovation, and collective will, we can protect our planet for current and future generations.

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Thank You

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